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CHAPTER 8

Nonresponse: Dealing with the Problem

Understanding survey nonresponse and its statistical implications leaves unresolved the question of how to deal with the problems it creates for analysis. In Chapter 7 we saw expressions for nonresponse bias when estimates are computed from nonrespondent data alone without any effort to compensate for the nonresponse. The size of these no-compensation biases under a deterministic view of nonresponse depends on two factors: the expected response rate and the degree to which respondents and nonrespondents differ according to values of the Y variable, the measure of substantive interest in the survey. Therefore, it is not surprising to find that most methods for dealing with nonresponse address either or both of these determinants.

In this chapter we explore some options available to the researcher for dealing with nonresponse. In many surveys several of these options are chosen as part of a general strategy for handling the missing data problem. The options might be described briefly as follows:

1. *Do Nothing.* The researcher elects to produce estimates based only on respondent data. Choosing this option indicates that either the researcher is unaware of the implications of nonresponse or believes them to be negligible. In the latter case the observed response rate may be sufficiently high or the likelihood of respondent–nonrespondent differences in the Y variable sufficiently low to merit the assumption that nonresponse bias will be near zero and thereby ignorable. In some surveys with high response rates this may be a workable option, although an absence of respondent–nonrespondent differences requires the usually naive assumption that respondents are a random subset of the survey sample.

2. *Reduce the Number of Nonrespondents.* Under this option, the researcher, mostly through improvements in the data collection strategy, increases the likelihood of obtaining useful data from members of the sample.

3. *Substitution.* New sample members in this approach are substituted for unit nonrespondents as a means to maintain the intended sample size, although the bias from nonresponse will not usually be reduced.

4. *Assess the Potential for Bias.* This method calls for an analysis of survey response or an investigation of the potential for respondent–nonrespondent differences, possibly involving the calculation of response rates for various domains or a comparison of respondents and nonrespondents on auxiliary data.

5. *Compensate for the Problem.* This option encompasses a broad grouping of methods where the researcher, in consideration of the nonresponse issue, makes some kind of alternation in the sampling design or in how the survey data are treated for analysis. The alternation may take any of the following forms:

a. *Extrapolation.* Estimates applicable to nonrespondents are projected by observing domain estimates from respondents ordered by degree of resemblance to nonrespondents.

b. *Nonrespondent Subsampling.* Final estimates are obtained using data from respondents and a subsample of nonrespondents.

c. *Weighting Adjustment.* Respondent data are adjusted to account for the nonresponse that has occurred, and the weighting adjustments are then used in the analysis. Although theoretically applicable to both unit and item nonresponse, adjustments are most often used to handle the first type of missing data. For simple estimators of means and totals, one type of weighting adjustments is equivalent to imputing the mean of similar respondents, thus making some applications of weighting adjustment methods an implicit form of imputation.

d. *Explicit Imputation.* This class of compensation procedures is applied when some data items obtained from unit respondents are either missing or found to be unusable. Following a planned process, each missing data item is replaced by a numerical value that is intended as a prediction of the missing value.

e. *Model-Based Inference.* A statistical model involving assumptions about nonresponse and the formation of the sampled population is at the heart of this class of compensation strategies. Parameters of the model are estimated using respondent data, and the fitted model is then used to predict data items for nonrespondents and ultimately, to obtain population estimates.

Some of the compensation methods can be distinguished by the assumptions made about differences between respondents and nonrespondents. Following the deterministic view of nonresponse, some presume that the means of all respondents ($\bar{Y}_1$) and nonrespondents ($\bar{Y}_0$) are the same. Others assume survey respondents to be a random subset of the selected sample, in

which case the expected value of survey measures, over outcomes of the sampling process, would be equal for respondents and nonrespondents. Finally, for some model-based methods, expected values of $\bar{Y}_1$ and $\bar{Y}_0$, over outcomes of the model, are presumed equal.

Our discussion of methods for dealing with the nonresponse problem generally follows the categories of options given above. We also distinguish between methods designed specifically for dealing with unit nonresponse and item nonresponse because the literature, despite the conceptual similarities between the two types of nonresponse, is largely segregated, for practical reasons. Methods for handling item nonresponse are those in which explicit efforts are made to improve completion rates for individual questionnaire items for respondents or where some well-defined process is followed to replace missing data items. All other methods presented in this chapter deal with the problem of unit nonresponse, although some (e.g., weighting class adjustments) could, although not very practically, be applied at the item level.

As in Chapter 7, our aim is to highlight and briefly explain some of the methods available in the literature. We give the reader a cursory view of what can be done to deal with nonresponse by reviewing material that we believe typifies available options. Expanded discussion on most topics is available through the many excellent articles and books cited in the text. In particular, we recommended the three-volume work by the Panel on Incomplete Data (Madow et al. 1983a, b; Madow and Olkin, 1983), Kalton (1983), and Cox and Cohen (1985).

8.1 UNIT NONRESPONSE

We first consider those approaches that have been designed to deal with nonresponse at the unit level, which in many surveys accounts for a larger portion of the missing data problem than does item nonresponse. The many and varying methods we review in this section generally fall into two classes; those where the goal is to avoid the problem by maximizing the unit response rate among sample members, and those where some form of compensation is used to reduce the effect on survey error of the unit nonresponse that ultimately occurs. Useful reviews on compensation methods have been produced by Chapman (1976), Kalton and Kasprzyk (1982), and Little (1982).

8.1.1 Prevention

Perhaps the best strategy for dealing with any problem is to keep it from becoming too large in the first place. Limiting bias associated with nonresponse requires steps to control either of its two determinants: respondent-nonrespondent differences and the response rate. Although somewhat affected by the response rate, the extent to which sample respondents and nonrespondents differ is difficult to control since the amount of difference

depends on which population members would be respondents or nonrespondents. Moreover, membership in these two groups depends on the relationship between a member's likelihood of responding and the measurement of interest.

Although it is usually unreasonable to expect that respondent-nonrespondent differences can be controlled directly, the response rate as the second determinant of bias can be regulated by introducing data collection strategies which reduce the likelihood that a population member, if selected in the survey, would be a nonrespondent. A number of these prevention methods are summarized from more comprehensive discussions given by Kish (1965), Warwick and Lininger (1975), and Mosteller (1978).

More formally, the goal of preventive methods is collectively, and in some instances selectively, to increase response probabilities (p_i) in the population. From a stochastic perspective this has the effect of increasing the expected number of respondents in the population, $N_1 = \sum_{i=1}^N p_i$, and the expected response rate, $\lambda_1 = N_1/N$, over the outcomes of the random response process at work in the population. In the deterministic view, increasing response probabilities means convincing some nonrespondents ($p_i = 0$) to become respondents ($p_i = 1$). This activity has the effect of increasing N_1 , the size of the respondent subclass, and $\lambda_1 = N_1/N$, the proportion of the population falling into it. Under both views the manifestation of λ_1 in the survey sample is the response rate, $l_1 = n_1/n$, where n is the sample size and n_1 is the number of respondents, which may be interpreted as either the number of sample members who respond as the result of the random response process operating on each sample member (stochastic view) or the number of respondent subclass members chosen in the sample (deterministic view).

While in principle one hopes for complete response in surveys, the realistic practitioner knows that this goal is usually unattainable. There then remains the matter of deciding what level of λ_1 is acceptable or, in more concrete terms, what is an acceptably high response rate. Perhaps the answer to this question cannot be given in absolute terms (e.g., 75 percent or higher as an acceptable response rate) but rather, is best couched in terms of several criteria. First the acceptability of a response rate may be tied to one's prior expectations in view of the survey topic, the population being studied, and the mode of data collection. A 75 percent response rate, for instance, would be considered quite acceptable in a self-administered mail survey of physicians on treatment practices but somewhat low for a personal interview survey of adults in the general population on some relatively innocuous health care issues. The topic, population, and mode of data collection in the latter survey all lend themselves to a greater likelihood of response than do the same three characteristics of the former survey.

Once the mode of data collection is established, the cost of steps to improve response rates is a second criterion that may have some bearing on deciding whether a response rate is acceptable. For example, an 80 percent

response rate for a personal interview survey might be considered acceptable if increasing the response rate to 85 percent would cause the per-unit cost of data collection to double because of increased travel and follow-up efforts. The complexity of steps that would have been required to improve the response rate in a given survey must also be considered in resolving the issue of whether a reported rate is good enough.

The acceptability of a reported response rate also depends on whether and what types of methods have been used to compensate for nonresponse. One good illustration may be taken from mail surveys, where a 50 percent response rate would be deemed unacceptable if nothing else were done to deal with the nonresponse. On the other hand, this rate might be viewed quite differently if nonrespondent subsampling (Section 8.1.4) or a weighting class adjustment procedure (Section 8.1.5) were used. These procedures would serve to eliminate at least partially the negative effects of the initial response level, thus placing the 50 percent rate in a somewhat more favorable light.

Finally, the substantive importance of findings from the study must be considered in judging reported rates of response. Response rates of 80 to 90 percent may be unnecessary in low-budget surveys designed, for example, to confirm that a strong front-runner in a political race will be elected or that a popular local referendum will pass. On the other hand, a rate exceeding 90 percent would be essential for any survey that had important implications for a multibillion dollar health care program.

This and our earlier discussion on defining relative measures of nonresponse (Section 6.2.2) point to the need for caution when using the response rate as a criterion for evaluating the quality of a survey. Clearly, one must look beyond this often misreported and potentially misleading number to determine the net effect of nonresponse on data obtained in the study. To put a reported response rate into a proper perspective, it is important to consider how the rate was defined, how amenable the population studied would have been to the survey topic, what mode of data collection was used, whether cost-effective measures were taken to maximize the rate, whether other nonresponse compensation procedures were used, and the impact of the study's findings.

The remainder of this section is devoted to a brief presentation of the preventive strategies often followed in practice. Although we tie these strategies to location, solicitation, and data collection efforts in different types of surveys, the material presented here should be viewed only as a starting point for the survey researcher who is devising a plan to increase the response rate. The reader is referred to texts by Moser and Kalton (1972), Warwick and Lininger (1975), and Dillman (1978) to supplement our discussion.

The basis for discussion in the remainder of this section is Table 8.1. To minimize redundancy among the lists we have omitted the following three obvious strategies, which apply to most situations covered in the table:

Table 8.1 Preventive Methods to Use (When Feasible) in Dealing with Unit Nonresponse

Point of Application	Mode of Data Collection	
	Mail Self-Reporting	Telephone/Personal Interview
Location	More current mailing address ^a High-priority mailing ^a	More call attempts ^a Clearer interview assignment materials ^a Improved call scheduling
Solicitation	Special tracing efforts ^a Follow-up reminders ^a Privacy and confidentiality assurances Incentives ^a Endorsements ^a Proxy respondents ^a Refusal conversion strategies ^a Miscellaneous enhancements ^a	Special tracing efforts ^a More call attempts ^a Privacy and confidentiality assurances Incentives ^a Endorsements ^a Use of female interviewers ^a Lead letter Reduced interviewer work load Refusal conversion strategies ^a
Data collection	Brief and concise interview Simplified questionnaire format Avoidance of sensitive subjects	Brief and concise interview Avoidance of sensitive subjects

^aAdditional discussion in text.

1. *Selective Recruiting.* Hire staff with experience in dealing with survey nonresponse. For example, one usually prefers interviewers with satisfactory past experience in doing surveys similar to the one for which they are being recruited. The reasons for such preference are obvious. These individuals are more likely to understand the problems faced in obtaining a response from sample members and to have found ways to handle these problems.

2. *Improved Training.* Manuals and sessions devoted to interviewer training include some discussion on how to deal with problems related to nonresponse. Overlooking this aspect of training forces the interviewer to improvise, which may be less effective.

3. *Improved Staff Supervision.* Those in a position to solicit participation in the survey are supervised more closely to assure that methods established to prevent nonresponse are applied correctly.

Several methods listed in Table 8.1 are discussed briefly.

8.1.2 Preventive Methods

8.1.2.1 High-Priority Mailing

A mailed questionnaire would have a greater chance of reaching a sample member if the package is sent by registered or certified mail, where someone must sign for it. Mailgrams or telegrams might also be effective in notifying someone of having been selected into a survey. Kernan (1971) and Dillman (1978) discuss the relative merits of various types of postage.

8.1.2.2 More Call Attempts

In interview surveys a *call attempt* refers to an event in which an interviewer tries to contact a selected member of the sample with the aim of ultimately completing an interview. Allowing a larger number of call attempts before considering a sample member to be a nonrespondent usually increases the chances of response, provided that calls are conveniently scheduled for sample members. With each additional attempt we are more likely to find available those who are often unavailable, and to find more willing those who in previous attempts may have been reluctant to participate in the study. Considering that increasing the allowable number of call attempts reduces nonresponse bias (and the corresponding mean square error of an estimated mean) while increasing survey costs, Deming (1953), Durbin (1954), and Thomsen and Siring (1983) use somewhat different approaches to determine the optimum number of call attempts.

8.1.2.3 Clearer Interviewer Assignment Materials

The primary linkage between the sampling and data collection operations is through the information the former provides the latter to determine who in the survey population is to be studied. When this information is unclear or vague, the chances increase that a sample member will eventually become an unlocatable nonrespondent. To avoid this problem, assignment lists given to field staff should include sufficient information to enable the interviewer to distinguish uniquely each sample member from among the rest of the population. Some items that help to clarify interviewer assignments are correctly spelled full name, complete street address with house number or description, home or business telephone number, and name and address of current employer. Maps sent with interviewers will further improve the process of location in personal interview surveys.

8.1.2.4 *Special Tracing Efforts*

In longitudinal surveys where some study subjects move between rounds or in cross-sectional surveys where the information available to locate sample members is incomplete or outdated, intensive efforts may be needed to find an address or telephone number. *Tracing*, the process of locating sample members in these situations, involves pursuing leads based on information available from one or more of the following: the sampling frame, a previously completed questionnaire, directory assistance, informants, and publicly available records. The chances of successfully locating subjects is clearly related to the amount of available tracing information and the skill with which it is used.

8.1.2.5 *Follow-up Reminders*

Some time after a questionnaire is first sent to each sample member in mail surveys, a follow-up in the form of a postcard, letter, telegram, or telephone call might be made to encourage those not having yet responded to do so. Several follow-up procedures and their effectiveness in reducing the extent of the nonresponse problem have been reviewed by various authors [see, e.g., Scott (1961), Hochstim and Athanasopoulos (1970), Dillman (1972), Linsky (1975), Kanuk and Berenson (1975), Duncan (1979), and Rao (1983a, b)].

8.1.2.6 *Incentives*

Because participation in some surveys requires a significant commitment of a respondent's time and energy, some sort of incentive payment to those agreeing to participate may be offered to respondents (e.g., money, a gift, a redeemable coupon, or an offer to send a report of the survey's findings). A number of studies [e.g., Dohrenwend (1970), Ferber and Sudman (1974), Chromy and Horvitz (1978), Gunn and Rhodes (1981)] generally point to improved response rates in most surveys if an incentive of some kind is offered to those sampled.

8.1.2.7 *Endorsements*

In some surveys, especially those involving professionals and institutions, endorsements of relevant societies and professional organizations are sought to enhance the credibility of the study, thereby reducing suspicion and increasing the likelihood of response. Although endorsements have been widely used, Kalsbeek and Hartwell (1977) have demonstrated a point of diminishing returns, regarding the number of endorsements, after a few key ones have been received.

8.1.2.8 *Use of Female Interviewers*

For some interview surveys female interviewers will tend to achieve higher response rates than male interviewers because women tend to arouse less suspicion and pose a less of a threat to certain segments of the population (e.g., the elderly). Although many accept this rationalization, there is little

strong evidence to suggest a relationship between response rates and the sex of the interviewer (Platek, 1977; Thompsen and Siring, 1983).

8.1.2.9 *Lead Letter*

A letter sent in advance of the initial contact in some interview surveys may enhance the credibility of the interviewer once the sample member is first contacted and thus increase the likelihood of response. This letter is best signed by someone whose name, title, or affiliation is held in high regard by members of the survey population. It should contain information designed to inform, assure, and motivate.

8.1.2.10 *Proxy Respondents*

A proxy respondent is a person who, although not selected in a survey sample, is considered a suitable substitute when the person actually selected cannot or should not participate. For example, in an interview survey any adult in the household who speaks English may be an allowable proxy for interviewing a non-English-speaking sample member. By using proxies nonresponse from inability on the part of the respondent will be less likely. On the other hand, use of proxies may contribute to increased measurement error since facts or opinions may be less accurately portrayed [see Roshwalb (1982)].

8.1.2.11 *Refusal Conversion Strategies*

Sample members refusing to participate based on an initial solicitation effort may in some instances be recontacted if it appears that there is some hope of converting them to respondents. The potential for conversion is not the same for everyone. Someone refusing because the interviewer called at a bad time might agree to participate if a convenient time could be found. A reluctant respondent might also be converted if the investigator can find a way to emphasize the study's credibility. On the other hand, someone emphatically expressing a distaste for surveys would be an unlikely participant under any circumstances.

Successfully converting a reluctant individual is tied to the reasons for refusal and to the investigator's ability to find arguments to refute these reasons. With this in mind, let us briefly examine how a conversion program might be set up in a survey. First, the investigator must attempt to learn *why* a potential respondent has initially refused for interview surveys. This can be done by training interviewers to probe gently for a reason by suggesting one for the respondent to confirm or revise (e.g., "Perhaps I've caught you at a bad time"). We have found that nonbelligerent refusers will often respond to this kind of probe by telling you why they are reluctant. In mail surveys it is difficult to learn who among nonrespondents are refusals, much less why they have refused. For that reason the conversion call must be by telephone or in person. Second, the investigator must develop an argument to deal with each possible reason for refusal and then find skilled interviewers to make the

conversions. Those assigned to convert refusals might, for example, be given a script containing hopefully effective counterarguments (e.g., reasons why the survey is important for those who think participating would be a waste of their time). In most instances conversion attempts are most effectively made a few days after the initial refusal, since employing a hard-sell strategy with someone at the time of an initial refusal is often counterproductive. The change in time may find the refuser in a more receptive frame of mind or, in household surveys, another person who will help pave the way for participation.

8.1.2.12 *Miscellaneous Enhancements*

Several other prevention methods may be used to increase the chances of soliciting participation in mail surveys. The utility of some of these methods has been tested. For example, Kahle and Sales (1978) have investigated the appearance of the envelope and Henley (1976) and Roberts et al. (1978) the use of deadlines for returning completed questionnaires, in each case concluding that such features may often improve response. These and similar studies point to the importance of sending attractive personalized material through the mail, of making a firm but reasonable request of the respondent's time, and of making it easy for the respondent to return the questionnaire. Scott (1961) and Dillman (1978) contain numerous suggestions along these lines.

8.1.3 Identification Studies

Another way of dealing with unit nonresponse is to identify the *potential* for nonresponse bias by attempting to determine whether the likelihood of response is somehow related to the major study variables, or equivalently, whether and how respondents differ from nonrespondents. On the basis of these so-called identification studies, the researcher can qualitatively assess the impact of nonresponse on survey estimates by investigating patterns of response rates among subgroups of the sample. Equivalently, these studies may also be done by comparing respondents with nonrespondents or respondents with the total population. Sociodemographic characteristics (e.g., age, ethnicity, gender, and education) available for responding and nonresponding sample members are commonly used.

In studying response-rate differentials among subgroups of the sample, the object is to determine the relationship between the likelihood of response and auxiliary variables thought to be highly correlated with the *Y* variable. Observing that subgroup response rates are statistically associated with values of the auxiliary variables may indicate that estimates obtained using the *Y* variable are biased because of nonresponse. This bias occurs because an association between response rates and the auxiliary variables points toward a nonzero correlation between individual response probabilities and values of the *Y* variable, and ultimately toward bias in survey estimates (see

Section 7.3). More intuitively, a relationship between response rates and the auxiliary variable implies that distributions of the Y variable and auxiliary variables among sample respondents are not the same as their corresponding distributions among members of the full sample. If the distribution of the Y variable among respondents is skewed because of nonresponse, estimates produced from the respondent sample will be skewed as well.

Identification studies are usually more telling for samples chosen from lists of individual population members than for samples chosen in clusters, especially geographic ones, because there is often more useful auxiliary information on lists. This kind of identification study could, for example, be done in a survey on access to health care conducted for a large company. Assuming that demographic data were available for each employee, response rates might be compared for various subgroups defined by age, race, and insurance coverage since all three would presumably be related to accessibility. Response-rate comparisons in area samples, on the other hand, are often limited to categories of qualitative characteristics for the area units within which the sample was chosen (e.g., population density, regional location).

Survey respondents and nonrespondents are compared using auxiliary data in a second type of identification study. Simple descriptive measures of the auxiliary variables (e.g., means) are typically computed in making these comparisons. The criteria for choosing the auxiliary variables are the same as in studies of subgroup response rates. In area samples the distributions by geographic area, population density, or observed racial or selected area units might be compared for respondents and nonrespondents. In other samples one might contrast some descriptive measure, or the distributions of various sociodemographic variables. To illustrate the latter case, consider the results of a study by Kalsbeek et al. (1974) designed to determine the impact of nonresponse on 17-year-olds selected in the National Assessment of Educational Progress (NAEP), a longitudinal survey of the educational competency of primary and secondary school children as determined by performance on standardized packages of test exercises. In this study respondents (students who showed up at scheduled testing sessions) were compared with nonrespondents (no-show students). A comparison of the mean and median for three auxiliary measures is presented in the top half of Table 8.2. Differences found in the table reveal that the respondents were generally better students, absent less often from classes, and taking a heavier course load. Assuming these measures to be reasonable correlates to test performance, we conclude that nontrivial respondent-nonrespondent differences on test performance are to be expected based on these data. Typical of the numerous other identification studies of the second type are those done by Mayer and Pratt (1966), Pucel et al. (1971), Gannon et al. (1971), Pavalko and Lukkerman (1973), and Settergen et al. (1983).

As in all studies of this kind the auxiliary information is used in lieu of data on the major study variables that are obviously unavailable for all nonrespondents. When the survey includes a subsampling of nonrespondents

Table 8.2 A Comparison of Responding and Nonresponding 17-Year-Olds in the National Assessment of Education Progress^a

	Respondents	Nonrespondents
Auxiliary Variables		
Grade point average ^b		
Mean	2.6	2.2
Median	2.7	2.2
Number of Courses taken during most recent semester		
Mean	5.4	4.7
Median	5.5	5.2
Percentage of school days absent in current year		
Mean	5.9	11.3
Median	4.2	7.3
Number of Correctly Answered Exercises^c		
Mathematics Package (total number of exercises)		
01 (16)	9.4	8.1
03 (19)	10.6	9.0
09 (19)	9.5	8.4
13 (22)	12.4	10.3
Science Packages		
01 (23)	12.0	10.7
03 (13)	6.2	5.6
09 (28)	18.9	17.9
13 (7)	3.2	2.9

^aThe response rate was 75 percent.

^bBased on a four-point scale (A = 4; B = 3; C = 2; D = 1; F = 0).

^cEntries in the table are weighted estimates of the average number of exercises answered correctly.

(Section 8.1.4), estimates applicable to nonrespondents based on the subsample can be compared with the corresponding estimates for respondents. Results of this kind of comparison for the NAEP study are presented in the lower half of Table 8.2. As suggested by the comparison using the auxiliary measures in the upper half of the table, we consistently find the estimated average number of correctly answered exercises to be higher for respondents than for nonrespondents.

The strategy in a third type of identification study is to look for indications of respondent-nonrespondent differences on the major study variables by comparing subgroups of respondents who differ according to the degree to which they are likely to resemble nonrespondents. This way of dealing with the nonresponse issue is sometimes used in mail surveys where data collection is completed in waves. The first wave is the period between the first mailing and first follow-up, the second wave is the interval between the first

and second follow-ups, and so on. The rationale for using waves is that sample members who respond in the c th wave indicate greater reluctance to participate (and thus stronger resemblance to nonrespondents) as c increases. For interview surveys a comparable measure to wave would be the number of call attempts needed to secure participation.

Respondent data and the wave or call attempt in which the response occurred have been used in different ways to identify and deal with the biasing effect of unit nonresponse. Most of the published accounts of this approach have been applied to mail surveys. In some instances trends observed from survey estimates obtained separately from each wave are noted [e.g., Dunkelberg and Day (1973) and Chapman (1976)], thus examining whether later respondents tend to differ from earlier respondents. Estimates using pooled data through each wave have also been compared for the same reason [e.g., Hawkins (1975) and Jones (1983)]. Assuming that the wave in which a person responds is a reasonable indication of likelihood to respond, monotonically increasing or decreasing trends among estimates by wave may indicate a correlation between individual response probabilities and the Y variable, thereby pointing toward biased estimates due to nonresponse.

Sequences of estimates, obtained by sorting respondents according to when during the period of data collection the response was obtained, may also be used to project estimates as if complete response had occurred. The term *extrapolation method* denotes a compensation procedure in which a sequence of estimates as described above is fitted to an assumed model, and the fitted model is used as the basis for extrapolation. The independent variable (x) in these models is some measure of the point during data collection when the questionnaire was completed (e.g., number of calls or time to completion, cumulative response rate), and the dependent variable (y) is a survey estimate (e.g., of a mean) obtained by pooling data from the start of data collection through the point determined by x .

The basic idea of most extrapolation methods is the same. Each respondent is assigned what we will call a *response outcome variable* (considered as an auxiliary X variable), indicating the degree to which the respondent is assumed to resemble a nonrespondent. An assumed model relating the response outcome variable to one of the major study variables (the Y variable) is then fitted using respondent data. Finally, the fitted model is used to extrapolate the final result. Approaches to extrapolation differ according to which measure is used for the X variable and which statistical model is assumed in relating the X variable to the Y variable.

In one of the first documented uses of the extrapolation idea applied to mail surveys, Hendricks (1949) fits a log-linear model,

$$y = \alpha x^{\beta}, \quad (8.1)$$

where x is the number of calls required to obtain the response, and where α

and β are constant regression parameters to be estimated from the respondent data. The definition of x Hendricks used is typical of the earlier extrapolation methods Scott (1961) summarized. Filion (1976) and Jones (1983) assume the linear model,

$$y = \alpha + \beta x, \quad (8.2)$$

where x is the cumulative response rate up to a given wave. Von Riesen and Novotny (1979) use time to completion for x and fit a piecewise linear model to reflect assumed changes in the linearity of the model during data collection. In mail surveys with one slope change this takes the form

$$y = \alpha + \beta_1 x + \beta_2 (x - X_t) \delta, \quad (8.3)$$

where X_t is the point on the x scale where the slope changes (when a follow-up reminder is mailed) from β_1 to $\beta_1 + \beta_2$ and

$$\delta = \begin{cases} 1 & \text{if } (x - X_t) \geq 0 \\ 0 & \text{if otherwise.} \end{cases}$$

Finally, Ognibene (1971), in a study that compares the feasibility of various models relating x (the cumulative response rate) to y , concludes that the hyperbolic model,

$$y = \alpha + \frac{\beta}{x}, \quad (8.4)$$

gives the most useful results.

The definition of x determines the manner in which the fitted model is used for extrapolation. When x is the cumulative response rate, the extrapolated estimate is for the Y variable in the case of complete response (where the cumulative response rate $x = 1$). However, when x is some measure of how much effort was involved in getting a response (wave, call attempt, or length of time to completion), the object of extrapolation is less clear. One must make some arbitrary choice of the response outcome variable that will adequately predict the value of the Y variable for nonrespondents.

The utility of extrapolation in dealing with survey nonresponse is not firmly established. Advocates of extrapolation would argue that the approach is a relatively simple and inexpensive way to deal with survey nonresponse. The results of extrapolation can be obtained without the expense of nonrespondent subsampling (Section 8.1.4) or complicated reweighting scheme (Section 8.1.5). On the other hand, detractors of extrapolation would question the strength of the relationship between measures of the time to

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completion and important study variables in light of existing evidence to the contrary. Chapman (1976), for example, finds no clear trends evident in a comparison of respondents obtained on successive call attempts in a large health care examination survey conducted by in-person interviewing. Scott (1961) also finds little evidence of a strong relationship when the Y variable is any one of several demographic characteristics, although he mentions some empirical findings which suggest that estimates obtained by extrapolation will sometimes be relatively good.

Despite some practical advantages, it would seem that extrapolation methods will not be widely used until measures of x that better predict values of the Y variable can be found. The problem with many measures is that they fail to represent the respondent's likelihood of participation. For example, the amount of effort required to obtain a response in an interview survey may reflect only the respondent's availability, not willingness or ability to participate. Presumably, an ideal response measure would reflect all three characteristics. In mail surveys, the wave in which response occurs and the time to completion may only indicate the respondent's tendency to procrastinate or forget.

8.1.4 Nonrespondent Substitution

In the event that unit nonresponse must not cause the final count of respondents to fall below the sample size originally intended for the study, one often-used method of dealing with the problem is to replace each nonrespondent in the initial sample with another member of the population, most frequently a member not included in the original sample. This technique, called substitution, is conceptually similar to explicit imputation used to deal with item nonresponse (see Section 8.2.2). We also note that when the substitute for a nonrespondent is chosen from among the respondents in the sample, substitution resembles a weighting adjustment procedure (see Section 8.1.5).

The rules followed in identifying the substitute respondent are generally random or nonrandom (Chapman, 1983). Random substitution rules require that the substitute be chosen by some type of probability sampling method, a feature that facilitates a statistical assessment of resulting identified estimates. In many applications the random substitute is selected from among members of the same subclass as the nonrespondent, where the subclass is defined in terms of variables thought to be correlated with the major study variables or within the same cluster in a cluster sample [see Waksberg (1978)]. The intent is for each nonrespondent and its substitute to be similar with respect to the variables used to define the subclass, and thus similar as to the major study variables. When substitution is done within subclasses, estimators from samples with substitutes have bias properties similar to those obtained from samples where a randomized hot-deck imputation (as discussed in Section 8.2.2) is used (Schaible, 1983).

Substitutes under a nonrandom designation scheme, on the other hand, are identified by applying (often during data collection) a predetermined set of criteria, but not by probability sampling. The substitute might, for example, be the dwelling unit immediately on the right as the interviewer faces the nonresponding dwelling in an area household sample, the next residential household found after adding one to the last digit of the nonrespondent's telephone number in a telephone survey, or more generally the next encountered member of the population with certain characteristics similar to those of the nonrespondent.

Two methods of dealing with survey nonresponse that are similar but not identical to substitution are worth noting. One is a technique suggested for repeated interview surveys by Kish and Hess (1959), in which a similar substitute is identified by random or nonrandom means from nonrespondents (after c call attempts) in a recent survey. Call attempts are then made to solicit participation from the substitute households. To increase the effectiveness of this replacement procedure, the nonrespondent in the current survey should have demographic characteristics similar to its substitute. This procedure may enable one to increase the response rate, since the investigator can afford to make more call attempts on the substitutes than on the sample nonrespondents, who will have had no more than c attempts made when replaced by a substitute with c attempts already made.

A second similar method to substitution is *supplementary sampling*, in which an initial sample and a set of smaller but randomly chosen supplementary samples are available to use in the event that nonresponse in the initial sample is higher than expected. This method is often used in one-time surveys in which there is relatively little prior knowledge of response rates in the population being studied. The procedure works as follows. Given that the expected response rate in the survey is thought likely to be no lower than λ_L but no higher than λ_H , an initial sample of size $n\lambda_H^{-1}$ is chosen. Following the same sampling design as the initial sample, a set of m supplementary samples of size $n(\lambda_L^{-1} - \lambda_H^{-1})/m$ is chosen independently. As the survey progresses, individual supplementary samples are added until the supplementary sample results in a sample size as close to n as desired. A full commitment must be made to obtaining participation from each member as each supplementary sample is added. Clearly, the larger m is allowed to be, the closer one can get to the desired sample size and the more supplementary sampling resembles individual substitution. Supplementary samples are most useful in surveys where making individual substitutions is too difficult or costly.

Substitution reduces nonresponse bias to the extent that the substitutes are collectively identical in every respect to the sample members for whom the substitutions are made. When substitution is done completely at random and the substitutes are subject to the same data collection protocol as the initial sample, nonresponse bias is unlikely to be substantially affected, since substitutes would replace nonrespondents with respondents resembling those

already in the sample (Kish, 1965, Section 13.6B). If, on the other hand, each respondent and its substitute can be matched on known correlates of the *Y* variable, the biasing effect of nonresponse will be diminished for the same reasons that weighting class adjustments (Section 8.1.5) and hot-deck imputation (Section 8.2.2) reduce bias. Unfortunately, as with most remedies for nonresponse, substitution does not completely eliminate its biasing effects. Empirical studies by Cohen (1955) and Williams and Folsom (1977) on the use of matched substitutes have confirmed this finding.

Some of the advantages and disadvantages of substitution may be summarized from the discussion by Chapman (1983). Among its advantages is that targeted sample sizes in matching categories can be achieved, thus controlling overall sample size and associated variance components from sampling error. Stratum sample sizes can also be controlled by matching within strata—thus, for example, avoiding variance estimation problems caused by strata with fewer than two sample members. A second advantage discussed earlier is that nonresponse bias may be reduced, though not eliminated, if nonrespondents and their substitutes are similar with respect to the major study variables. Among the drawbacks to substitution is that the availability of a substitute may diminish the intensity of an interviewer's efforts to obtain a response, knowing that a backup is "waiting in the wings." The result would be a lower response rate for the initial sample than would be expected if no substitutes were available. Thus every effort should be made to obtain a response from members of the initial sample before substitution is allowed. A second drawback of substitution is the potential for overstating the survey response rate by including substitutes in the numerator but excluding the substituted nonrespondents from the denominator of the calculated rate. A more reasonable rate would be calculated from the initial sample alone or from the initial sample plus those designated as substitutes (and either responding or not).

8.1.5 Nonrespondent Subsampling

At the root of the problem of nonresponse is the availability of survey data for respondents and a corresponding lack of data for nonrespondents. One highly visible class of nonresponse compensation procedures was developed with the idea that estimates which are free of nonresponse bias can be obtained by using data from the sample of respondents and a random subsample of initial nonrespondents, to whom more intensive, costly efforts are applied to obtain a response. Nonrespondent subsampling procedures are most often used in mail surveys, where response rates are low and where more intensive efforts (e.g., by telephone or in person) will obtain useful responses from a high percentage (although usually not 100 percent) of a nonrespondent subsample.

The original idea of nonrespondent subsampling, suggested by Hansen and Hurwitz (1946), follows the classical approach to statistical inference.

Their idea has given rise to numerous extensions and revisions, some of which use model-based approaches to inference. Most of the existing theory presumes that responses are obtained from all subsampled nonrespondents. This provision seldom holds completely, since despite extraordinary efforts, some hard-core nonrespondents will remain, such as those adamantly opposed to surveys in general or those who are unavailable during the entire survey.

The approach followed in all classical nonrespondent subsampling methods is as follows. Consider the survey population to consist of $H \geq 2$ mutually exclusive and exhaustive strata. One of these strata contains all the nonrespondents, while the other $H - 1$ strata are distinguished by the intensity of efforts (e.g., number of call attempts, follow-up letters, etc.) required to obtain a response.

In their original article, Hansen and Hurwitz assume that $H = 2$ and use double sampling or two-phase sampling to select those in the nonrespondent stratum. The population of size N consists of N_1 respondents and $N_0 = N - N_1$ nonrespondents. With simple random sampling (without replacement) used in selecting the initial sample (of size n with n_1 initially responding) and the nonresponding subsample, the estimator of the mean $\bar{Y}$ using the Hansen-Hurwitz method is

$$\bar{y}_{HH} = l_1 \bar{y}_1 + (1 - l_1) \bar{y}_{01}, \quad (8.5)$$

where $l_1 = n_1/n$ is the response rate in the initial sample, $\bar{y}_1 = \sum_{i=1}^{n_1} y_i/n_1$ is the mean among initial respondents, and $\bar{y}_{01} = \sum_{i=1}^{n_{01}} y_i/n_{01}$ is the mean among the n_{01} subsampled nonrespondents. The variance of the Hansen-Hurwitz estimator, given the assumed design, will be

$$\text{Var}(\bar{y}_{HH}) = \frac{(1 - f)[S^2 + (g - 1)\lambda_0 S_0^2]}{n}, \quad (8.6)$$

where $f = n/N$, $\lambda_0 = N_0/N$, $g = n_0/n_{01}$, and S^2 and S_0^2 are variances of the Y variable among all population members and all members of the nonrespondent stratum, respectively. Assuming that $S^2 = S_0^2$, we note that Eq. (8.6) reduces to $\text{Var}(\bar{y}_{HH}) = (1 - f)S^2/n'$, where $n' = n/[1 + (g - 1)\lambda_0]$. Since $n' \leq n$ we see that $\text{Var}(\bar{y}_{HH})$ is greater than or equal to the variance of the estimated mean from a simple random sample of size n , assuming complete response. In other words, $\bar{y}_{HH}$ may remove the bias from nonresponse, but its variance will always be greater than the complete-response variance unless all nonrespondents are subsampled ($g = 1$).

Subject to a simple cost model, the optimum rate of nonrespondent subsampling (g^{-1}) can be determined by choosing that value of g which minimizes $\text{Var}(\bar{y}_{HH})$ for fixed expected cost. Allowing for a few simplifying

assumptions, this optimum value is

$$g_{\text{opt}} = \left[\frac{C_{01}(S^2 - \lambda_0 S_0^2)}{(C^* + \lambda_1 C_1) S_0^2} \right]^{1/2}, \quad (8.7)$$

where $\lambda_1 = 1 - \lambda_0$, and C^* , C_1 , and C_{01} are average per-unit costs for choosing and setting up the initial sample, gathering data for the initial respondents, and gathering data for the nonrespondent subsample, respectively. In the case where one can reasonably assume that $S^2 = S_0^2$ and that C^* is small relative to $\lambda_1 C_1$, we see that $g_{\text{opt}}^{-2} \doteq C_1/C_{01}$, indicating that the optimum subsampling rate is inversely related to the relative increase in the unit cost of obtaining and processing data from the nonrespondent subsample.

As an alternative to g_{opt} , Srinath (1971) suggests an optimum value for g that does not rely as heavily on knowing λ_1 accurately. The optimum subsample size under the alternative rule is computed as

$$n_{01}^* = \frac{n_0^2}{ng_{\text{opt}}^* + n_0}, \quad (8.8)$$

where

$$g_{\text{opt}}^* = \frac{C_{01}\lambda_0^2(S^2 - \lambda_0 S_0^2)}{(C^* + \lambda_1 C_1) S_0^2} - \lambda_0 = (g_{\text{opt}} - 1)\lambda_0. \quad (8.9)$$

The advantage of the Srinath solution is that regardless of what we know about λ_0 before the survey, g_{opt}^* will yield an estimator with the desired precision by allowing the subsampling fraction to vary depending on the response rate in the sample. Using g_{opt} in Eq. (8.7), the precision of estimates may depart from targeted levels when the actual and assumed values of λ_0 differ.

Several extensions that broaden the applicability of nonrespondent subsampling are mentioned, based on more extensive discussions by Zarkovich (1966), Cochran (1977), and Rao (1983a). El-Badry (1956) considers the case where multiple attempts are made through follow-up letters or calls before a nonrespondent subsample is chosen, and the data obtained through each attempt and the subsample are combined to produce the final estimate. Under this framework $H > 2$ strata are defined, one for nonrespondents (which is subsampled) and $H - 1$ strata for the respondents to each of the $H - 1$ attempts made to obtain a response. Rao (1968) considers nonrespondent subsampling with $H = 2$ but where the list from which the sample is drawn contains an unknown number of duplicated entries. Rao (1973) has also applied nonrespondent subsampling to the setting where stratification is used in choosing the initial sample. Finally, Rao and Hughes (1983) present an optimized solution for nonrespondent subsampling applied to mail surveys

when the object is to estimate the difference of means between two domains that may not be defined by strata used for sampling. Two alternative sampling schemes are considered in which nonrespondents are subsampled twice.

Bartholomew (1961) has suggested an idea somewhat related to nonrespondent subsampling for studies in which a maximum of two call attempts are allowed. The selected sample thus consists of three distinct groups: respondents after the first call (group R_1), respondents after the second call (group R_2), and nonrespondents (group NR). Assuming that group R_2 is a fair representation of groups R_2 and NR combined, group R_2 is treated as if it were a nonrespondent subsample and the data from it reweighted to produce an estimator comparable to $\bar{y}_{HH}$ in Eq. (8.5), where l_1 is the proportion of the sample in the group, R_1 , $\bar{y}_1$ is obtained from R_1 , and $\bar{y}_{01}$ is obtained from group R_2 . Clearly, the reduction in nonresponse bias from this approach is directly related to how collectively similar members of groups R_2 and NR are.

We turn next to a series of articles in which Bayesian methods are used to obtain estimates when nonrespondent subsampling has been done. Bayesian statistical inference is based on sample data (as in the classical approach) but also on expressions of prior knowledge concerning parameters to be estimated. Applications of these methods are often characterized by the need for two samples: a pilot sample to establish or substantiate prior assumptions about certain important study parameters and to aid in planning subsequent survey sampling, and the main sample where design plans are implemented. We mention these Bayesian methods here, although a later section is devoted entirely to these model-based methods.

The pioneering work by Erickson (1967) is useful to illustrate the Bayesian approach to nonrespondent subsampling, since his work is a direct adaptation of the Hansen-Hurwitz approach. Essential features related to sampling and data collection are the same in both approaches. The steps followed in obtaining estimates from the survey differ markedly, however. Beginning with joint prior distributions for the proportion of respondents in the universe (λ_1) and for the true mean for respondents (μ_1) and nonrespondents (μ_0), Erickson obtains a joint posterior distribution of $(\mu_1, \mu_0, \lambda_1)$ through the first phase of sampling from likelihood functions for μ_1 and λ_1 , given observed data from the first-phase pilot sample of n_1 respondents. An overall joint posterior distribution for $(\mu_1, \mu_0, \lambda_1)$ is then obtained, given data from the combined two-phase sample. The final step consists of establishing the posterior distribution for the overall mean, $\mu = \lambda_1\mu_1 + (1 - \lambda_1)\mu_0$, from the preceding step. From this posterior distribution of μ an optimized Bayes estimator ($\hat{\mu}_E$) can be derived by minimizing the expectation of the loss function,

$$L = K(\mu - \hat{\mu}_E) + C^*n + C_1n_1 + C_{01}n_{01}, \quad (8.10)$$

over the posterior distribution of μ , given the initial sample and nonrespondent subsample, where K is a constant. In the special case where prior information is vague, it can be shown that $\hat{\mu}_E$ is equivalent to the Hansen-Hurwitz estimator. Optimum values for n_{01} and n are also derived.

Several other uses of Bayesian methods in surveys with nonrespondent subsampling have been suggested. For example, in their Bayesian predictive approach to double sampling, Rao and Ghangurde (1972) suggest the selection of two two-phase samples, a pilot sample and a main sample. Both sampling designs call for less expensive efforts to be used in obtaining response from an initial simple random sample, followed by more expensive efforts to obtain data from a nonrespondent subsample. Assuming a similar design framework, Singh and Sedransk (1978b) consider a Bayesian approach to regression analysis in the presence of unit nonresponse. These same authors [in Namboodiri (1978)] also use Bayesian methods to estimate the population mean. This work is extended to the case where a stratified two-phase sample is used [in Singh and Sedransk (1983)].

To conclude, consider the following summary of the differences between classical and Bayesian approaches to the use of nonrespondent subsamples:

Criterion	Classical	Bayesian
1. Basis for inference	Sampling distribution of the $n_1 + n_{01}$ respondents determined by methods used to select initial sample and subsequent ones	Prior assumptions concerning behavior of key parameters expressed as a stochastic model
2. Optimality criteria for n and subsampling rates (g^{-1})	Minimizing variance of estimator (based on its sampling distribution), gives prespecified cost	Minimizing an expected loss function determined by cost and some measure of difference between parameter and its estimator
3. Basis of choice of n and g	Assumed variance and cost models as well as reasonable measure of parameters associated with these models	Assumed prior distribution and data from pilot sample

8.1.6 Adjustment Procedures

Strategies for dealing with unit nonresponse described to this point have been those which the investigator invokes at points in the study other than when the major study analyses are being done. Preventive methods, nonrespondent substitution, and nonrespondent subsampling are applied earlier in the study, during the period of data collection. Identification studies are often done as a preliminary step in analysis or as part of an evaluation of the study protocol after analysis has been done. Methods described in this section and the next are those the investigator uses in the process of formulating and producing survey estimates.

We first look at strategies dealing with how the rate of nonresponse varies somewhat within most survey samples and thus affects the sample's ability to present a balanced cross section of the survey population. Consider the following example to illustrate this point. In the absence of nonresponse, an equal-probability sample for a survey of adults in the United States would proportionately represent urban and rural residents. If nonresponse occurred and, as is typically the case, urban response rates were lower than rural response rates, the final set of respondents would include a disproportionately large share of rural adults. The classical methods discussed next deal with the imbalance from nonresponse by adjusting the respondent data, in effect, by computing a numerical weighting factor which when applied to each respondent's data at least partially returns a balance to the respondent sample.

Since the concept of adjustment is important for us here, some discussion of its meaning in the present framework is in order. First we must establish what is to be adjusted. To do so, consider the simple problem of estimating a total (Y), where we know that the estimator Horvitz-Thompson $\hat{Y}_{HT} = \sum_{i=1}^n W_i Y_i$, when applied to probability sample of size n , will be unbiased. In the absence of nonsampling error the sample weight, $W_i = \pi_i^{-1}$, where π_i is the sampling probability, adequately reflects the likelihood that data from the i th member of the sample would be used for estimation. When nonresponse is present and there are data for $n_1 < n$ sample members, the estimator, $\hat{Y} = \sum_{i=1}^{n_1} W_i Y_i$, will no longer be unbiased (Section 7.3). Unbiasedness is possible if we use a revised weight that accounts for the selection probability (π_i) and the conditional probability that the i th sample member would, if selected, respond and provide useful data for estimation, that is, its response probability (p_i), which must exceed zero for all members of the population. Thus, to achieve unbiasedness, the Horvitz-Thompson estimator must be modified to $\hat{Y}_{HT}^* = \sum_{i=1}^{n_1} W_i^* Y_i$, where $W_i^* = (\pi_i p_i)^{-1}$ is a nonresponse-adjusted sample weight. Nargundkar and Joshi (1975) present some of the theory of this estimator in the absence of other nonsampling errors, as do Platek and Gray (1983) with measurement errors present (see Section 7.3). The estimator $\hat{Y}_{HT}^*$ has been adjusted in the sense that response probabilities for the respondents are used to account for the nonresponse in the survey.

This last statement motivates the underlying principle of nonresponse adjustment procedures in classical survey inference. To minimize the biasing

effect of nonresponse, a measure of each sample member's response probability is used to adjust sample weights. Since the p_i are unknown, they must be estimated in some reasonable manner. The way in which p_i is estimated distinguishes the various adjustment methods.

In one of the first known adjustment methods, attributed to Politz and Simmons (1949) but based on an idea by Hartley (1946), it is necessary to obtain t_i , the number of days during the previous five that the i th respondent would have been available to be interviewed at the same time of day. The estimated response probability for the Politz-Simmons method is then $\hat{p}_i^{(1)} = (t_i + 1)/6$ (see Section 7.2).

A second adjustment method, called the *weighting class adjustment*, estimates response probabilities by dividing the original sample (including both respondents and nonrespondents) into H mutually exclusive and nonoverlapping subsets, called *adjustment cells*, labeled using the subscript h , within which members are assumed to have similar values of the Y variable and all response probabilities are presumed to be equal. Since the entire sample is partitioned, it is apparent that information used to make assignments must be available for respondents and nonrespondents. The type of homogeneity desired for adjustment cells is similar though less restrictive than the homogeneity sought for the same cells if they were used for sampling. For this nonresponse adjustment we wish for $\bar{Y}_{1h} = \bar{Y}_h$ and $S_{1h}^2 = S_{0h}^2$, while for minimizing sampling error we try to form homogeneous cells to serve as sampling strata (we wish for $S_h = 0$; $h = 1, 2, \dots, H$). The latter homogeneity criterion is, however, desirable for both hot-deck and cold-deck imputation methods of dealing with item nonresponse (Section 8.2.2).

The generalized estimator of p_i used in the weighting class adjustment method is a weighted within-cell response rate obtained as follows. For any probability sample where $W_{hi} = \pi_{hi}^{-1}$ is the unadjusted sample weight for the i th member of the h th adjustment cell, the estimator of p_{hi} (the subscript h must be added to the index adjustment cell) will be $\hat{p}_{hi}^{(2)} = \sum_{i=1}^{n_{1h}} W_{hi} / \sum_{i=1}^{n_h} W_{hi}$, and the nonresponse adjustment for the i th sampled unit in the h th cell is $1/\hat{p}_{hi}^{(2)} \geq 1$, where n_h is the number of sample members in the h th cell and n_{1h} is the number of respondents in the h th cell. The weighting-class-adjusted weight is thus $W_{hi}^{(2)} = W_{hi} / \hat{p}_{hi}^{(2)}$. When simple random sampling or any other equal probability design is used to choose the original sample, $\pi_{hi} = n/N$, for all $h = 1, 2, \dots, H$ and $i = 1, 2, \dots, n_h$, and $\hat{p}_{hi}^{(2)} = n_{1h}/n_h$.

A third approach to adjusting for nonresponse is the poststratification adjustment method. In the absence of nonresponse, poststratification enables one to reduce the sampling variance of estimates by adjusting sampling weights within each of H adjustment cells (Hansen et al., 1953a). The poststratification adjustment for each weight in the h th adjustment cell is computed here as

$$a_h = \frac{N_h \sum_{h=1}^H \sum_{i=1}^{n_h} W_{hi}}{N \sum_{i=1}^{n_h} W_{hi}}, \quad (8.11)$$

where measures of N_h , the number of population members in the h th adjustment cell or the size of N_h relative to $N = \sum_{h=1}^H N_h$, are obtained from the best data available (e.g., a recent census). For an equal probability sampling design the adjustment simplifies to $a_h = N_h n / (N n_h)$. We also note that a_h tends to be closer to unity as n_h increases and $\sum_{i=1}^{n_h} W_{hi} / \sum_{h=1}^H \sum_{i=1}^{n_h} W_{hi}$ improves as an estimator of N_h/N .

When used to deal jointly with sampling errors and nonresponse bias, poststratification and weighting class approaches can be combined to form the cell-level adjustment for W_{hi} ,

$$a_h^* = \frac{N_h \sum_{h=1}^H \sum_{i=1}^{n_h} W_{hi}}{N \sum_{i=1}^{n_{1h}} W_{hi}}. \quad (8.12)$$

For simple random sampling in the initial sample, $a_h^* = N_h n / (N n_{1h})$. We see from Eqs. (8.11) and (8.12) that $a_h^* \geq a_h$, with equality possible only if there is complete response within cells. The combined adjustment can be rewritten as $a_h^* = a_h / \hat{p}_{hi}^{(2)}$, from which we observe that the measure used to estimate response probabilities would be $\hat{p}_{hi}^{(3)} = \hat{p}_{hi}^{(2)} / a_h$, although a_h^{-1} may be more appropriately conceived as an adjustment to π_{hi} than as a modification to an estimate of p_{hi} . Thus the adjusted sample weight under the combined adjustment scheme is computed as $W_{hi}^{(3)} = a_h^* W_{hi}$, which, when equal probability sampling has been used, simplifies to $W_{hi}^{(3)} = N_h / n_{1h}$.

To this point we have assumed that the same adjustment cells are defined to use the weighting class and poststratification methods together. In practice, however, quite different cells may be used for each adjustment. When adjustment cells are defined separately for the two adjustments, perhaps one of the most commonly used approaches to dealing with unit nonresponse is first to compute a weighting-class-adjusted weight for a respondent in the h' th weighting class adjustment cell as $W_{h'i}^{(2)} = W_{h'i} / \hat{p}_{h'i}^{(2)}$, and then to compute the poststratification adjustment for the h th poststratification adjustment cell as

$$a_h^* = \frac{N_h \sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(2)}}{N \sum_{i=1}^{n_{1h}} W_{hi}^{(2)}}, \quad (8.13)$$

where $W_{hi}^{(2)}$ is the value of the weighting-class-adjusted weight for a respondent who is a member of the h th poststratification adjustment cell. The final combined weight used in analysis is $W_{hi}^{(4)} = a_h^* W_{hi}^{(2)}$, with a_h^* computed from Eq. (8.13).

From the description above of the procedure for computing weighting class and poststratification adjustments, we note one important difference in practical requirements. The weighting class adjustment requires two items of information for respondents and nonrespondents: unadjusted weights (W_{hi}) and the measures for each of the variables used to define the adjustment

cells. The poststratification adjustment [a_h^* in Eq. (8.12)] requires an external measure of N_h and measures of the adjustment cell variables, but only for respondents.

Cohen and Kalsbeek (1981) give an example of the common practice of combining weighting class and poststratification adjustments for the National Medical Care Expenditure Survey. In this study, weighting class adjustment cells were census enumeration districts and block groups, which served as second-stage sampling units. Cells for the poststratification adjustment, computed using weighting-class-adjusted weights, were 32 age-race-sex cross-classification categories as adjustment cells. Bailer et al. (1978), as well as Rizvi (1983a) and Madow (1983), give other example of the joint application of the two adjustments.

A few other methods based on the idea of identifying reasonable measures of p_i are mentioned briefly. Chapman (1976) discusses one where in the p_i are estimated from regression analysis. Drew and Fuller (1980, 1981), in their strategy combining callbacks and adjustment cells, estimate p_{hi} , $\Delta_h = N_h/N$, and the proportion of hard-core nonrespondents from a fitted multinomial model. Thomsen and Siring (1983) likewise use weighting classes to fit a model relating p_i to the number of callbacks. As for all adjustment methods, the ability to reduce nonresponse bias is directly related to how well p_i can be estimated.

For simple estimators of means and totals the weighting class adjustment method can be viewed as an implicit form of imputation where the imputed value (Z_i) is inferred rather than a value of the Y variable for one of the respondents (Section 8.2.2). The linkage between these two types of compensation procedures is explored a bit further using Fig. 8.1, which illustrates the weighting class method within an adjustment cell consisting of six respondents and two nonrespondents. All unadjusted sample weights (W_{hi}) among units within the cell are assumed to be equal. In step 1, consider each of the eight units to consist of six equal-sized parts with the unit's Y variable associated with each part. The measure used by each unit in analysis is the simple average of the Y variable associated with the six parts. Since the Y variable is unavailable for nonrespondents U_7 and U_8 , step 2 of the weighting class method calls for $Y_1, Y_2, \dots, Y_6$ to be "weighted up" by a factor of 8/6 (the weighting class adjustment). This procedure is equivalent to dividing the nonrespondents' parts among the respondents and producing estimates with six rather than eight datum points coming from this cell. Since each unit consists of equal-sized parts, we see in step 3 that weighting up is equivalent to a redistribution of parts such that the simple average of the Y variable among respondents has effectively replaced its original value for the nonrespondents, U_7 and U_8 . Thus we see in this case that applying the weighting class adjustment is the same as imputing the mean among respondents in the cell. We note also that the sample size for producing estimates from weighting-class-adjusted data would be the number of respondents (n_1) rather than the full sample size (n).

Step	Respondents						Nonrespondents	
	U1	U2	U3	U4	U5	U6	U7	U8
(1) The problem	Y_1 Y_1 Y_1 Y_1 Y_1 Y_1	Y_2 Y_2 Y_2 Y_2 Y_2 Y_2	Y_3 Y_3 Y_3 Y_3 Y_3 Y_3	Y_4 Y_4 Y_4 Y_4 Y_4 Y_4	Y_5 Y_5 Y_5 Y_5 Y_5 Y_5	Y_6 Y_6 Y_6 Y_6 Y_6 Y_6	Y_7 Y_7 Y_7 Y_7 Y_7 Y_7	Y_8 Y_8 Y_8 Y_8 Y_8 Y_8
(2) The weighting-class adjustment applied	Y_1 Y_1 Y_1 Y_1 Y_1 Y_1	Y_2 Y_2 Y_2 Y_2 Y_2 Y_2	Y_3 Y_3 Y_3 Y_3 Y_3 Y_3	Y_4 Y_4 Y_4 Y_4 Y_4 Y_4	Y_5 Y_5 Y_5 Y_5 Y_5 Y_5	Y_6 Y_6 Y_6 Y_6 Y_6 Y_6		
(3) Implicit imputation of the respondent mean	Y_1 Y_1 Y_1 Y_1 Y_1 Y_1	Y_2 Y_2 Y_2 Y_2 Y_2 Y_2	Y_3 Y_3 Y_3 Y_3 Y_3 Y_3	Y_4 Y_4 Y_4 Y_4 Y_4 Y_4	Y_5 Y_5 Y_5 Y_5 Y_5 Y_5	Y_6 Y_6 Y_6 Y_6 Y_6 Y_6	Y_1 Y_2 Y_3 Y_4 Y_5 Y_6	Y_1 Y_2 Y_3 Y_4 Y_5 Y_6

Figure 8.1 Relationship between weighting class adjustment and imputation (equal sample weights within adjustment cell).

The amount of survey error associated with both explicit and implicit imputation methods depends on how much imputed values differ from their corresponding true values (in Fig. 8.1 on how much the respondent mean differs from Y_7 and Y_8). The difference will be small when the adjustment cell is homogeneous with respect to the Y variable. The amount of homogeneity depends, in turn on the level of statistical association between the Y variable and the variables used to define the adjustment cell.

In the remainder of this section we discuss separately the Politz-Simmons weighting class and poststratification adjustment methods in the context of analyses where means and totals are being estimated. We will not consider the effectiveness of these adjustment procedures on other more complex estimates, although Santos (1981) has examined the effects of imputing adjustment cell means when estimating more complex parameters.

8.1.6.1 Politz-Simmons Adjustment

For the case where a total (Y) is to be estimated from a complete enumeration, Politz and Simmons (1949) demonstrate some properties for the estimates, $\hat{Y}_{PS} = \sum_{i=1}^N Y_i / \hat{p}_i^{(1)}$, where $\hat{p}_i^{(1)} \equiv p_i = (t_i + 1)/6$ and t_i is the number of times during the past five days that the i th respondent would have been available to be interviewed at the same time of day. The bias and variance of $\hat{Y}_{PS}$ are found to be

$$\text{Bias}(\hat{Y}_{PS}) = - \sum_{i=1}^N (1 - p_i)^6 Y_i \quad (8.14)$$

and

$$\text{Var}(\hat{Y}_{PS}) = \sum_{i=1}^N Y_i^2 \left\{ 6 \sum_{j=1}^6 \frac{p_{ij}}{j} - [1 - (1 - p_i)^6]^2 \right\}, \quad (8.15)$$

respectively, where for the i th population member's p_i is a response probability (with *response* equated to *availability*), and p_{ij} is the probability that the member would be found at home j times if call attempts were made on six consecutive days at the same time each day.

We see from Eq. (8.14) that the Politz-Simmons estimator is not unbiased. $\hat{Y}_{PS}$ will always tend to underestimate Y unless $p_i = 1$ for all population members; however, the absolute value of the bias relative to Y will be small [i.e., less than $\bar{Y} - 1$, since $\sum_{i=1}^N (1 - p_i)^6 \leq N$].

The estimator was originally designed to circumvent the cost of multiple call attempts. Adjusting the data with $\hat{p}_i^{(1)}$ was thought to be as useful as attempting to move response probabilities closer to unity by additional call attempts. One of the approach's limitations is that $\hat{p}_i^{(1)}$, while dealing with nonresponse from nonavailability, fails to account directly for nonresponse linked to refusal, although the value of t_i given by each respondent may to some extent reflect willingness and availability to participate. This point is relevant because when $\hat{p}_i^{(1)}$ is unrelated to the Y variable, the variance of $\hat{Y}_{PS}$ is needlessly high and its bias no lower than if nothing had been done to deal with the nonresponse. The approach also fails to account for those who would not have been available at all during the six-day period (those with $p_i = 0$). These hard-core nonrespondents have no chance of being represented

in the sample and thus, given the schedule of call attempts, can never be represented in the survey. For additional discussion of the Politz-Simmons adjustment method, see Deming (1953), Zarkovich (1966), Kalton (1983), and Cochran (1977), Cochran presents properties of the Politz-Simmons estimator of a mean.

8.1.6.2 *Weighting Class Adjustment*

The cells defined to produce weighting class adjustments in multistage designs are often sampling units from one of the first stages of sampling. The rationale for defining cells in this way is that individuals living in geographic proximity are relatively more similar than the population as a whole, and that information on sampling unit affiliation is known for both respondents and nonrespondents. For example, the minor civil divisions selected in the first stage of a three-stage sample of households might be designated as the adjustment cells. Perhaps for this reason, Platek et al. (1977) refer to the criteria that define the cells for weighting class procedure as "design-dependent" balancing factors.

In unclustered list samples adjustment cells might be formed from one or more data items (variables) available on the sampling frame. For example, in a survey of physicians' record-keeping practices where the sample is selected from a membership listing of the American Medical Association, the variables used to form weighting class adjustment cells might be the physician's specialty and size of practice. In some instances these cells are defined by variables used earlier in the survey for stratified sampling. In other instances the sampling strata are also used as adjustment cells.

The ideal set of variables would be those strongly associated with the major study variables but mutually unrelated. It also seems preferable to define cells by a coarse division on several acceptable variables than to form the same number of cells by a finer division on the best single variable among those considered acceptable [see Kish and Anderson (1978)]. The choice of variables is often somewhat arbitrary, although more sophisticated cluster analysis techniques might be used [e.g., see Hartigan (1975)]. Since we hope for a high level of association between the variables used to form cells and the major study variables, which in turn are often related to the p_i , it is not uncommon and is even desirable to find observed within-cell response rates (l_{1h}) to vary considerably among cells. In picking the number (H) of cells to form, one is torn between two competing influences. On one hand, a large number of well-formed cells will more effectively reduce bias, but the wider variation in adjusted sample weights caused by more variable weighting class adjustments will cause the variance of estimates to increase. Thus, keeping H small is likely to reduce the variance, but increasing H is likely to reduce the bias.

To examine some of the statistical properties of estimates using the weighting class adjustment, we consider the case where the object is to estimate the population mean $\bar{Y}$. For any probability sample with known

sample weights $W_{hi} = (\pi_{hi})^{-1}$, the estimator can be expressed as

$$\begin{aligned}\hat{Y}_{WC} &= \sum_{h=1}^H \hat{\Delta}_h \hat{Y}_{1h} \\ &= \frac{\sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(2)} Y_{hi}}{\sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(2)}},\end{aligned}\quad (8.16)$$

where $\hat{\Delta}_h = \sum_{i=1}^{n_h} W_{hi} / \sum_{h=1}^H \sum_{i=1}^{n_h} W_{hi}$ estimates $\Delta_h = N_h / N$, $\hat{Y}_{1h} = \sum_{i=1}^{n_{1h}} W_{hi} Y_{hi} / \sum_{i=1}^{n_{1h}} W_{hi}$ estimates $\bar{Y}_{1h} = \sum_{i=1}^{N_{1h}} Y_{hi} / N_{1h}$, the mean among all respondents in the h th cell, and

$$W_{hi}^{(2)} = W_{hi} \frac{\sum_{i=1}^{n_h} W_{hi}}{\sum_{i=1}^{n_{1h}} W_{hi}}. \quad (8.17)$$

When $H = 1$, a single survey-wide adjustment is computed. Since this adjustment is a constant over n_1 respondents, $\hat{Y}_{WC} = \sum_{i=1}^{n_1} W_i Y_i / \sum_{i=1}^{n_1} W_i$, meaning that the unadjusted and weighting-class-adjusted estimators are equivalent.

For the special case where simple random sampling is used in choosing the initial sample, $\hat{\Delta}_h = \delta_h = n_h / n$, $\hat{Y}_{1h} = \bar{y}_{1h} = \sum_{i=1}^{n_{1h}} Y_{hi} / n_{1h}$, and $W_{hi}^{(2)} = N n_h / n n_{1h} \propto l_{1h}^{-1}$ with $l_{1h} = n_{1h} / n_h$. Thus $\hat{Y}_{WC} = \sum_{h=1}^H \delta_h \bar{y}_{1h}$ and when $H = 1$, $\hat{Y}_{WC} = \bar{y}_1 = \sum_{i=1}^{n_1} Y_i / n_1$. Kalton (1983) calls the weighting class adjustment in this setting a *sample weighting adjustment* and presents its basic properties. The bias of $\hat{Y}_{WC}$ given by Kalton but shown earlier by Thomsen (1973) is

$$\text{Bias}(\hat{Y}_{WC}) = \sum_{h=1}^H \Delta_h \lambda_{0h} (\bar{Y}_{1h} - \bar{Y}_{0h}) \quad (8.18)$$

and its variance is

$$\begin{aligned}\text{Var}(\hat{Y}_{WC}) &= \frac{\sum_{h=1}^H \Delta_h S_{1h}^2 / \lambda_{1h} + \sum_{h=1}^H \lambda_{0h} S_{1h}^2 / \lambda_{1h}^2 n + \sum_{h=1}^H \Delta_h (\bar{Y}_{1h} - \bar{Y}_1^*)^2}{n},\end{aligned}\quad (8.19)$$

where S_{1h}^2 is the element variance among all respondents in the h th cell and $\bar{Y}_1^* = \sum_{h=1}^H \Delta_h \bar{Y}_{1h}$. We see from Eq. (8.18) why forming homogeneous adjustment cells is important. Complete homogeneity with respect to the Y variable among respondents and nonrespondents in each cell implies that $\bar{Y}_{1h} = \bar{Y}_{0h}$, which in turn implies that $\bar{Y}_{WC}$ will be unbiased.

In the event that adjustment cells are formed but that the unadjusted (no-compensation) estimator, $\bar{y}_1 = \sum_{i=1}^{n_1} Y_i / n_1$, is used to estimate $\hat{\bar{Y}}$, Kalton has also noted that the difference between the bias of $\bar{y}_1$ and $\hat{\bar{Y}}_{WC}$ is

$$\text{Bias}(\bar{y}_1) - \text{Bias}(\hat{\bar{Y}}_{WC}) = \sum_{h=1}^H \frac{\Delta_h(\bar{Y}_{1h} - \bar{Y}_1)(\lambda_{1h} - \lambda_1)}{\lambda_1}. \quad (8.20)$$

The difference in bias therefore depends on properties of the adjustment cells. If the cells were formed so that $\lambda_{1h} = \lambda_1 (h = 1, 2, \dots, H)$, $\bar{Y}_{1h} = \bar{Y}_1 (h = 1, 2, \dots, H)$, or more generally that the respondent means ($\bar{Y}_{1h}$) and expected response rates (λ_{1h}) were uncorrelated among cells, there will be no difference in the bias of $\hat{\bar{Y}}_{WC}$ and $\bar{y}_1$. Moreover, when the cells are not completely homogeneous, we cannot be assured that $\hat{\bar{Y}}_{WC}$ will have a smaller bias than $\bar{y}_1$. For example, we see from Eqs. (8.18) and (8.20) that when $\bar{Y}_{1h} > \bar{Y}_1 (h = 1, 2, \dots, H)$ and λ_{1h} and $\bar{Y}_{1h}$ are inversely related among all cells, $\text{Bias}(\bar{y}_1) < \text{Bias}(\hat{\bar{Y}}_{WC})$. In more practical terms these results warn us that it is possible to do more harm than good by using weighting class adjustments. A similar comparison of the variances of $\bar{y}_1$ and $\hat{\bar{Y}}_{WC}$ by Kalton does not point to any clear preference between the two estimators, although when $\bar{Y}_{1h} = \bar{Y}_1$ and $S_{1h}^2 = \bar{S}_1^2 (h = 1, 2, \dots, H)$,

$$\text{Var}(\hat{\bar{Y}}_{WC}) - \text{Var}(\bar{y}_1) = \bar{S}_1^2 \sum_{h=1}^H \frac{\Delta_h / \lambda_{1h} - \Delta_{1h} / \lambda_1}{n}, \quad (8.21)$$

where $\Delta_{1h} = N_{1h} / N_1$. The variance difference in Eq. (8.21) is nonnegative, with the increase in the variance of $\hat{\bar{Y}}_{WC}$ over that of y_1 because of variation in the final weights $[W_{hi}^{(2)}]$ brought about by the variation among weighting class adjustments.

Comparisons of bias and variance for $\bar{y}_1$ and $\hat{\bar{Y}}_{WC}$ lead to an interesting but troublesome paradox when useful adjustment cells can be formed. Cells with wide variation in λ_{1h} contribute to bias reduction in choosing $\hat{\bar{Y}}_{WC}$ over $\bar{y}_1$ but a variance increase since variable λ_{1h} yields variable l_{1h} , which produce variable weights, since $W_{hi}^{(2)} \propto l_{1h}^{-1}$. Therefore when choosing to use $\hat{\bar{Y}}_{WC}$, we must hope that our adjustment cells will lead to $\text{MSE}(\hat{\bar{Y}}_{WC}) \leq \text{MSE}(\bar{y}_1)$, where in general $\text{MSE}(\cdot) = \text{Var}(\cdot) + \text{Bias}^2(\cdot)$.

Platek and Gray (1983) present an illustration of the weighting class adjustment procedure in the presence of errors from sampling, measurement, and nonresponse (from a stochastic viewpoint). Assuming that N is known, the estimator of $\bar{Y}$ would be

$$\hat{\bar{Y}}_{PG} = \sum_{h=1}^H \sum_{i=1}^{n_h} \frac{R_{hi} W_{hi}^{(2)} V_{hi}}{\pi_{hi} N}, \quad (8.22)$$

where for the i th member of the sample in the h th adjustment cell $W_{hi}^{(2)} = W_{hi}(n_h/n_{1h})$, $V_{hi} = Y_{hi} + \varepsilon_{1hi}$ is the value of Y_{hi} observed with measurement error equaling ε_{1hi} , and

$$R_{hi} = \begin{cases} 1 & \text{if the } h\text{th sample member responds} \\ 0 & \text{if otherwise} \end{cases}$$

is the response status indicator. Assuming that adjustment cells have been predefined and that the adjustment is applied independently in each cell, the bias of $\hat{Y}_{PG}$ is

$$\text{Bias}(\hat{Y}_{PG}) = \sum_{h=1}^H \sum_{i=1}^{N_h} \frac{p_{hi} B_{1hi}}{\bar{p}_h} + \sum_{h=1}^H \sum_{i=1}^{N_h} \frac{(p_{hi} - \bar{p}_h) Y_{hi}}{\bar{p}_h}, \quad (8.23)$$

where $B_{1hi} = E_t(\varepsilon_{1hi})$ is the expected individual measurement error (Section 7.3) and $\bar{p}_h = \sum_{i=1}^{n_h} p_{hi}/n_h$. The first component of Eq. (8.23) might be viewed as bias because of measurement error, while the second is bias attributed to within-cell correlation between response probabilities and the Y variable. The rather involved variance expression for $\hat{Y}_{PG}$ consists of the following five additive components, attributed to:

1. Sampling
2. Measurement error arising from ε_{1hi}
3. The covariance between sampling and measurement error
4. The stochastic nature of nonresponse arising from R_{hi}
5. The covariance between R_{hi} and $R_{hj}(i \neq j)$ within adjustment cells

Although the weighting class adjustment method could conceivably be used to deal jointly with unit and item nonresponse, this option is seldom chosen since separate adjustments would be required for each data item. This can be seen by noting that the set of n_{1h} respondents at the item level would often differ in size and composition if "respondent at the item level" were defined as a sample member who participates in the survey and provides an answer to the item. When the sets of n_{1h} respondents within adjustment cells differ among items, the corresponding weighting class adjustments, $\sum_{i=1}^{n_h} W_{hi} / \sum_{i=1}^{n_{1h}} W_{hi}$, are likely to differ as well, thus requiring separate adjustments and final weights $[W_{hi}^{(2)}]$ for each item. This procedure is cumbersome and creates problems for any analysis involving combinations of items (e.g., cross-tabulations, regression modeling), which may explain why

weighting class adjustments are used almost exclusively as a method to deal with unit nonresponse. The same point holds for the weighting class adjustment used in combination with poststratification.

Kalton (1983) investigates the properties of a version of the weighting class adjustment method for $H = 1$, which is implicitly hot-deck imputation (Section 8.2.2) applied at the unit level. This approach to dealing with nonresponse was first discussed by Hansen et al. (1953b) and Kish (1965) and will be called the randomized reweighting method, which when $n_1 \geq n_0$ ($l_1 \geq 0.5$) works as follows. An equal probability subsample of size n_0 is selected with or without replacement from the n_1 respondents. Considering the case where there is one survey variable with missing data, each subsampled respondent is matched to a particular nonrespondent and becomes the donor unit for the nonrespondent, since the respondent's Y variable is imputed implicitly for the nonrespondent's Y variable. Imputation is implicit since application of the method calls for the sample weight of the nonrespondent to be added to the weight of its "donor," although properties of estimators when the imputation is explicit are the same [see Oh and Scheuren (1983)]. The respondent's (donor's) weight is incremented (or adjusted) by the amount equaling the nonrespondent's weight. Unlike the usual weighting class adjustment where (for $H = 1$) we assume that $\bar{y}_1 = \bar{y}_0$, in the randomized reweighting method we have $E_s(\bar{y}_0) = \bar{y}_1$, where expectation is over all samples of size n with n_1 respondents.

The distinction between the usual weighting class adjustment and randomized reweighting can be seen by examining the reweighting scheme more generally (by not requiring that $l_1 \geq 0.5$) when some equal-probability method is used to select the initial sample. Let $I = \text{Tr}(l_1^{-1})$ denote $l_1^{-1} = n/n_1$ truncated to the nearest integer. Then $n_1^* = n \pmod{n_1}$ respondents in the h th adjustment cell are assigned the adjustment $I + 1$, and $(n_1 - n_1^*)$ respondents are assigned the adjustment I . By contrast, the adjustment assigned to all respondents under the weighting class procedure would be $l_1 - 1$ such that $I \leq l_1^{-1} < (I + 1)$.

Historically, randomized reweighting is a throwback to the earlier approach where computer punch cards for respondents were randomly duplicated to compensate for nonresponse and there was no reweighting of the data. This approach, conceptually identical to randomized reweighting, produces estimates with the same nonresponse bias as the survey-wide nonresponse adjustment (the weighting class adjustment with $H = 1$). Unfortunately, estimates using randomized reweighting have been shown to have less desirable variance properties than those of estimates using the survey-wide adjustment (Kalton, 1983). This may partially explain why duplication of respondent records is seldom used in present-day surveys.

The extent to which the variance of the estimator using the reweighting is greater than the variance of the estimator, $\bar{y}_1 = \sum_{i=1}^{n_1} Y_i/n_1$, which uses a survey-wide nonresponse adjustment, will depend on how the subsample of respondents is chosen in the randomized reweighting method. The more

efficient the subsampling design (e.g., using stratification), the lower the amount of variance increase (Kalton and Kish, 1984).

Provided that $n_1 \geq n_0$ and that simple random sampling (without replacement) is used to choose the initial sample and the respondent subsample, the randomized reweighting estimator of $\bar{Y}$ is $\hat{\bar{Y}}_{RR} = (n_1 \bar{y}_1 + n_0 \bar{y}_d)/n$, where $\bar{y}_d$ is the mean of the n_0 respondent values assigned as donors. The proportionate increase in variance, conditioned on fixed n_1 , is

$$\frac{\text{Var}(\hat{\bar{Y}}_{RR}) - \text{Var}(\bar{y}_1)}{\text{Var}(\bar{y}_1)} = (1 - l_1)(2l_1 - 1), \quad (8.24)$$

where $l_1 = n_1/n \geq 0.5$ and the maximum proportionate increase of 0.125 occurs when $l_1 = 0.75$.

Platek and Gray (1983) present properties of a randomized reweighting scheme applied to adjustment cells and where errors associated with sampling, nonresponse (following a stochastic view), and measurement are considered. They assume that the population consists of H mutually exclusive and exhaustive adjustment cells, within which the cell total on the Y variable (Y_h) is estimated as

$$\hat{Y}_h = (I_h + 1) \sum_{i=1}^{n_{1h}^*} \frac{R_{hi} V_{hi}}{\pi_{hi}} + I_h \sum_{i=1}^{n_{1h} - n_{1h}^*} \frac{R_{hi} V_{hi}}{\pi_{hi}}, \quad (8.25)$$

where $(n_{1h} - n_{1h}^*)$ respondents in the h th cell are assigned the adjustment $I_h = \text{Tr}(I_{1h}^{-1})$, and n_{1h}^* are assigned the adjustment $(I_h + 1)$. Assuming that the number of population members in the h th cell (N_h) is known, the population total and mean for the Y variable can be estimated as $\hat{Y}_{PG}^* = \sum_{h=1}^H \hat{Y}_h^*$ and $\hat{\bar{Y}}_{PG}^* = \hat{Y}_{PG}^*/N$, respectively. The expected value of $\hat{Y}_{PG}^*$ from the random weighting method as described will be equivalent to the expected value of $\hat{\bar{Y}}_{PG}$ under the usual weighting class adjustment [see Eq. (8.22)], where the expectation for each estimator is taken over all stochastic sources of variation affecting each estimator. The variance of $\hat{\bar{Y}}_{PG}^*$ (or $\hat{Y}_{PG}^*$) consists of the same components as the variance of $\hat{\bar{Y}}_{PG}$ plus five similar components arising from the randomized subsampling of respondents to assign the adjustments, I_h or $I_h + 1$.

8.1.6.3 Poststratification Adjustment

Adjustment cells used for poststratification are formed in much the same way as strata for sample selection; however, they are defined by variables not available at the time the original sample was selected. They are always mutually exclusive and exhaustive subsets of the sample and one hopes that values of the Y variables in each cell are more similar than among all values in the sample. Since poststratification adjustment cells are formed after the

survey has been done, the variables used to define the cells are often included in the survey questionnaire. The best poststratification variables are those strongly correlated to the Y variable. As a result, they are often correlated with individual response probabilities (if a stochastic view is taken), since the Y variable and the response probability for population members are usually correlated. Thus as with weighting class adjustment cells, we expect members of poststratification adjustment cells to have similar response probabilities.

Use of poststratification in the National Health Interview Survey (NHIS) conducted by the National Center for Health Statistics (Kovar and Poe, 1985) is fairly typical. In this survey of the civilian, noninstitutionalized population in the United States, each respondent is assigned to one of 60 age-race-sex cross-classification cells for which reliable current population figures $\Delta_h = N_h/N$ are available independent of the survey. A poststratification adjustment is computed for the h th cell ($h = 1, 2, \dots, 60$) as

$$a_h^* = \Delta_h \frac{\sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(2)}}{\sum_{i=1}^{n_{1h}} W_{hi}^{(2)}},$$

where $W_{hi}^{(2)}$ is the raw sample weight ($W_{hi} = \pi_{hi}^{-1}$) times a weighting class adjustment. The final adjusted sample weight is $W_{hi}^{(3)} = a_h^* W_{hi}^{(2)}$, from which it can be shown that $\sum_{i=1}^{n_{1h}} W_{hi}^{(3)} / \sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(3)} = \Delta_h$. Thus we see in general that using a poststratification adjustment forces the weighted relative frequency distribution among cells to correspond precisely to the relative distribution among those same cells in the population. By using this adjustment the NHIS sample weights are in essence finally adjusted to bring the sample "into line" with the U.S. population, at least with respect to the joint distribution by age, race, and sex as defined in the 60 cells. A sample distorted by nonresponse, poor sample coverage, and sample variation now has weights allowing the weighted data more accurately to estimate parameters whose measurement (values of the Y variable) is correlated to the three poststratification variables.

Hansen et al. (1953a), Cochran (1977), and Oh and Scheuren (1983) make a number of practical suggestions regarding the formation of adjustment cells for poststratification. Several apply to the weighting class approach as well. One is that all cells should be large enough so that all n_{1h} are sufficiently great (say, 20 or more). Related to the importance of maintaining reasonable levels of n_{1h} is Thomsen and Siring's suggestion (1983) that gains from poststratification diminish as the number of adjustment cells (H) increases. The size constraints on n_{1h} and H point to the importance of considering cell-level estimates ($\bar{y}_{1h}$) when forming cells. Second, since the estimator is a function of $\Delta_h = N_h/N$, it is important for the measures of N_h to be reasonably accurate. Although perfect measures are seldom available, the measures used for N_h should be of higher quality than estimates of N_h that

could be obtained directly from the survey data. Third, adjustment cells should be internally homogeneous and externally heterogeneous with respect to the Y variable. For that reason adjustment cells are sometimes determined by clustering algorithms which minimize the variance of the Y variable within cells [e.g., Morgan and Sonquist (1963) and Gillo and Shelley (1974)]. Fourth, the variables used for poststratification are always categorical (or categorized) and should all be highly correlated with the Y variable but not with each other. Adjustment cells that are internally homogeneous with respect to highly correlated poststratification variables will also be internally homogeneous with respect to the Y variable. Finally, in the case where one has several suitable poststratification variables from which to choose, it is usually preferable to form cells by crossing a few levels on all variables than to pick the best one and divide it into numerous levels.

Consider next the special case, where the initial sample is chosen by simple random sampling, the same adjustment cells are used for the weighting class and stratification adjustments, and the mean ($\bar{Y}$) is to be estimated. Kalton (1983) presents statistical properties of the estimator that uses the weighting class and poststratification adjustments, namely,

$$\begin{aligned}\hat{Y}_{P-S}^* &= \sum_{h=1}^H \Delta_h \bar{y}_{1h} = N^{-1} \sum_{h=1}^H N_h \frac{y_{1h}}{n_{1h}} \\ &= \frac{\sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(3)} Y_{hi}}{\sum_{h=1}^H \sum_{i=1}^{n_{1h}} W_{hi}^{(3)}},\end{aligned}\quad (8.26)$$

where $\bar{y}_{1h} = \sum_{i=1}^{n_{1h}} Y_{hi}/n_{1h} = y_{1h}/n_{1h}$ and $W_{hi}^{(3)} = N_h/n_{1h}$. First note the distinction between $\hat{Y}_{P-S}^*$ and two other estimators: (1) the poststratification estimator in the absence of nonresponse, $\hat{Y}_{P-S} = \sum_{h=1}^H \Delta_h \bar{y}_h$, where $\bar{y}_h = \sum_{i=1}^{n_h} Y_{hi}/n_h$, and (2) the estimator $\hat{Y}_{WC} = \sum_{h=1}^H \delta_h \bar{y}_{1h}$, using the weighting class adjustment only. We might reasonably expect that $\hat{Y}_{P-S}^*$ is the best of the three, although it makes the unrealistic assumption that there is complete response in the survey. In presuming the presence of nonresponse, the estimator $\hat{Y}_{P-S}^*$ uses an adjustment that is the product of the weighting class and poststratification adjustments. It assumes that there reasonable measures for the Δ_h are available for the population and compensates for problems caused by an inadequate sampling frame (see Chapters 3 through 5) as well as nonresponse (e.g., in random-digit telephone surveys of the general population where some households do not have telephones and some of those that do would choose not to respond). Finally, $\hat{Y}_{WC}$ uses data available from the sample alone to compensate for nonresponse. Since δ_h , a sample estimator of Δ_h , is used in $\hat{Y}_{WC}^*$, we expect it to be subject to greater variation than $\hat{Y}_{P-S}^*$ and therefore less preferred.

One also notes from the first line of Eq. (8.26) the relationship between $\hat{Y}_{P-S}$ and the separate ratio estimator of the mean presented by Hansen et al. (1953a) and discussed further by Cochran (1977, Sec. 6.10). The auxiliary X variables in the separate ratio estimator may be likened to a "count" variable which equals 1 for all members of the population. The sample data on the Y variable and X variable used to produce the estimate are limited to those who respond. To complete the analogy between $\hat{Y}_{P-S}^*$ and the separate ratio estimator, we must assume for the former that respondents in each adjustment cell can be treated as a simple random subsample of those originally chosen.

Given the foregoing setting and that $E_s(\delta_h) = \Delta_h$, it is not surprising to find that

$$\text{Bias}(\hat{Y}_{P-S}^*) = \text{Bias}(\hat{Y}_{WC}) = \sum_{h=1}^H \Delta_h \lambda_{0h} (\bar{Y}_{1h} - \bar{Y}_{0h}), \quad (8.27)$$

which implies that, as with $\hat{Y}_{WC}$, the amount of nonresponse bias can be reduced to the extent that cells with equal respondent and nonrespondent means are formed. The variance of $\hat{Y}_{P-S}^*$ can be expressed as

$$\text{Var}(\hat{Y}_{P-S}^*) \doteq \text{Var}(\hat{Y}_{WC}) - \left[\frac{\sum_{h=1}^H (1 - \lambda_{1h}) S_{1h}^2}{\lambda_{1h}^2 n^2} + \frac{\sum_{h=1}^H \Delta_h (\bar{Y}_{1h} - \bar{Y}_s)^2}{n} \right], \quad (8.28)$$

where $\bar{Y}_s = \sum_{h=1}^H \Delta_h \bar{Y}_{1h}$. Since the biases of $\hat{Y}_{P-S}^*$ and $\hat{Y}_{WC}$ are equivalent and the bracketed term subtracted in Eq. (8.28) is greater than zero, we see that the variance and mean square error are always smaller for $\hat{Y}_{P-S}^*$ than for $\hat{Y}_{WC}$. This indicates that poststratification and the weighting class procedures applied to the same adjustment cells appear to complement each other in improving estimates, since when used in combination estimates are better than when the weighting class method is used alone. This finding then confirms our prior expectations. The reader is referred to Kalton (1983), the Panel on Incomplete Data (National Research Council, 1983), Oh and Scheuren (1983), and Thomsen and Siring (1983) for further discussion of the statistical and practical implications of the weighting class and poststratification approaches.

In general, we have seen that the adjustment methods operate by modifying the original sample weight ($W_i = 1/\pi_i$), which may vary because of the sampling design. The adjusted weight is computed as $W_i^* = W_i/\hat{p}_i$, where $\hat{p}_i$

as the estimator of the response probability for the i th sample member is likely to vary among survey respondents. Except for those instances where W_i and $\hat{p}_i$ are positively correlated, the variation in adjusted weights will exceed the variation of the original sample weights, since the $\hat{p}_i$ vary among respondents. The effect of added variation caused by the adjustments is to increase the variance of survey estimates.

One way to control the variation of adjusted weights is to smooth weights by setting upper and lower limits on the adjustments. For example, Banks et al. (1983) report setting an upper bound of 2 and a lower bound of 0.5 on the poststratification adjustments used in a 1970 national survey on health service use and expenditures in the United States.

In some instances, the excess from truncated weights will be spread over the sample, as seen in the following simple example. Suppose that for a set of n weights (W_i), we specify that all weights exceeding W' are to be trimmed and smoothed. Weights for the n_1 respondents with $W_i > W'$ will have an excess ($\varepsilon_i = W_i - W'$) which must be spread over the sample so that the sum of weights over the n_1 respondents is the same before and after smoothing. One approach to spreading the excess would be to adjust each of the n_1 weights by the factor $\phi = \sum_{i=1}^{n_1} W_i / (\sum_{i=1}^{n_1} W_i - \sum_{i=1}^{n_1} \varepsilon_i) \geq 1$, in which case the largest possible weight after smoothing will be $\phi W'$. If the smoothed weights must not exceed W' , the excess can be smoothed iteratively, with the excess of each iteration smoothed over the remaining set of untruncated weights.

Truncating adjusted weights can also be done by pooling the data. This practice is followed in *The Current Population Survey* [see U.S. Bureau of the Census (1978)]. Weighting class adjustments exceeding 2.0 are smoothed by pooling counts of respondents and nonrespondents in the exceptional adjustment cell with counts in other cells with similar demographic characteristics. In the unlikely event that the pooled adjustment factor exceeds 3.0, it is set equal to 3.0. For example, an adjustment cell with 20 households and a response rate of 25 percent (implying a nonallowable adjustment of $a_h = 4.0$), when combined with a group of cells with 100 households and an overall response rate of 95 percent, would yield an allowable $a_h = 1.2$ for all respondents in the pooled cells.

Variation in sample weights can also be controlled by what has been called the *raking-ratio adjustment* or *iterative proportional-fitting method*. First suggested by Deming and Stephan (1940), the object of the raking method is to circumvent some of the instability of adjustments produced from cross-classifications of two or more adjustment variables by adjusting weights so that marginal distributions rather than joint distribution of the variables obtained external to the survey are maintained. The procedure assumes a log-linear model relating adjustment cell response rates to marginal effects of the adjustment variables.

As evident from this brief description, raking methods require only that the marginal frequencies for all raking variables be known. That joint frequencies are not required turns out to be a practical advantage of the

method since marginal frequencies are often the only ones known or for which reasonable approximations are available.

Operationally, raking is similar to poststratification on two or more variables in that a single adjustment is the end product for each respondent. It differs from poststratification in that the process of computing each raking-ratio adjustment is iterative rather than a single computation. As with poststratification and the weighting class method, one presumes that probabilities within adjustment cells are equal. Moreover, rules for identifying adjustment variables and forming adjustment cells are the same as with the other two methods. In some instances raking might be used in combination with the other two methods. Oh and Scheuren (1978) present a bibliography of the use of raking methods.

Oh and Scheuren (1983) also describe a general protocol for raking when adjustment is over two variables, one with H categories, the other with K categories. If we let n_{1hk} ($h = 1, 2, \dots, H; k = 1, 2, \dots, K$) denote the number of respondents in each cell of the cross-classification, and let W_{hki} be the weight to be adjusted for the i th respondent in each cell, then we start with a weighted sum, $\sum_{i=1}^{n_{1hk}} W_{hki}$, in each cell. If N_{hk} denotes the true population count for the hk th cell, the marginal frequencies $N_{h.} = \sum_k N_{hk}$ and $N_{.k} = \sum_h N_{hk}$ are assumed to be known (or at least well approximated) from auxiliary sources. Through an iterative process beginning with a seed value ($a_{hki}^{(0)}$), an adjustment (a_{hki}^*) is computed for each weight such that $\sum_{h=1}^H \sum_{i=1}^{n_{1hk}} a_{hki}^* W_{hki}$ and $\sum_{k=1}^K \sum_{i=1}^{n_{1hk}} a_{hki}^* W_{hki}$ are acceptably close to $N_{.k}$ and $N_{h.}$, respectively, for all $h = 1, 2, \dots, H$ and $k = 1, 2, \dots, K$. For example, starting with $a_{hki}^{(0)} = 1$ and $W_{hki}^{(0)} = W_{hki}$, we might compute for the v th iteration the raking-ratio adjustment,

$$a_{hki}^{(v)} = \frac{N_{h.} N_{.k}}{\left[\sum_{k=1}^K \sum_{i=1}^{n_{1hk}} W_{hki}^{(v-1)} \right] \left[\sum_{h=1}^H \sum_{i=1}^{n_{1hk}} W_{hki}^{(v-1)} \right]}, \quad (8.29)$$

where in general $W_{hki}^{(v)} = W_{hki}^{(v-1)} a_{hki}^{(v-1)}$ ($v = 1, 2, \dots$). At each iteration the differences between the marginal weighted sums, $\sum_{h=1}^H \sum_{i=1}^{n_{1hk}} a_{hki}^{(v)} W_{hki}^{(v)}$ and $\sum_{k=1}^K \sum_{i=1}^{n_{1hk}} a_{hki}^{(v)} W_{hki}^{(v)}$, and the marginals for the auxiliary data, $N_{.k}$ and $N_{h.}$, respectively, would be computed to see whether adjusted weights have converged to the auxiliary marginals. If prespecified convergence criteria are eventually met on the V th iteration, the final raking adjustment for W_{hki} is $a_{hki}^* = \prod_{v=1}^V a_{hki}^{(v)}$, and the final weight for analysis is $W_{hki}^* = a_{hki}^* W_{hki}$. As we see, then, raking is named for the r , the scaling of unadjusted weights, with the extent and direction of scaling (up or down) dependent on how the weighted relative frequency of each cell stacks up in producing some known marginal frequencies.

A cursory assessment of the raking-ratio adjustment technique reveals some limitations. One problem is that convergence of adjusted marginals to the auxiliary marginals is not assured and little is presently known about

under what conditions convergence will occur. Much remains to be learned about this adjustment approach before it will be widely used, although several authors have already investigated the properties of estimates using raking-ratio adjustments [see, e.g., Brackstone and Rao (1981), Konijn (1981), and Kalton (1983)].

8.1.7 Bayesian and Other Model-Based Methods

Model-based methods are distinguished from classical methods in that true observable measures of the variable of interest (the Y variable) are considered to be the result of an assumed stochastic process, as opposed to being treated as fixed quantities. In effect, the vector $\mathbf{Y}' = (Y_1, Y_2, \dots, Y_N)$ is a vector of random variables in model-based methods and a vector of fixed constants in classical methods.

In this section we focus our attention on ways in which the model-based approach to inference has been used in dealing with the nonresponse problem. We note in our discussion that the model in model-based approaches to nonresponses assumes a stochastic mechanism for nonresponse. The stochastic form specified for this mechanism is used to produce estimators that compensate for the nonresponse. We also noted that probability sampling, though not necessary for many model-based methods, is advisable, since it can produce samples suitably balanced in their representation of the population sampled. Quota sampling is thought by some to be sufficient because of the balance it produces through quota controls (King, 1983).

Stochastic models used to explain the outcome of $\mathbf{Y}$ may take several forms. Some models assume the Y_i to be random variables with common moments but make no specific assumptions concerning the shape of the distribution for Y_i . In other models the Y_i are assumed to follow some particular distribution function, $f(Y_i; \theta_i)$, with an unknown but estimable L -dimensional parameter vector, $\theta'_i = (\theta_{i1}, \theta_{i2}, \dots, \theta_{iL})$. In most applications the Y_i are assumed to be mutually independent. Bayesian methods use these kinds of parametric models and add further prior stochastic assumptions concerning the parameters of the assumed distribution. In another type of model one assumes that the Y_i are related to a $(K + 1)$ -dimensional vector of auxiliary measures $\mathbf{X}'_i = (X_{i0}, X_{i1}, \dots, X_{ik})$ through some linear regression model such that $E_n(Y_i) = \beta'_i \mathbf{X}_i$, where $E_m(\cdot)$ denotes expectation over outcomes of the model and $\beta'_i = (\beta_{i0}, \beta_{i1}, \dots, \beta_{ik})$ is a vector of unknown regression coefficients, thus implying zero expectation for random error attributable to the model. Any of the models above might be *exchangeable* if the joint distributional function for all possible permutations of $\mathbf{Y}$ is symmetric in its N terms.

As mentioned in Section 7.3, many model-based methods that accommodate the nonresponse problem use respondent data combined with assumptions about the joint distribution explaining the origin of $(\mathbf{Y})$, the vector of stochastic response outcomes $(\mathbf{R})$, and the vector of outcomes $(\mathbf{D})$ from some